// Day 30 of 30

COMPLETE ROADMAP

Spring Boot Project Roadmap — 0 to Production

→ Swipe through all 7 slides

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Phase 1: Project Setup

Day 1 — Foundation

```
1. start.spring.io
   → Spring Web, JPA, MySQL, Security
   → Lombok, DevTools, Actuator

2. Project structure:
   controller/
   service/
   repository/
   model/
   dto/
   config/
   exception/
```

💡 Good structure = good project. Always.

Phase 2: Database + Entities

Day 2-3 — Data layer

```
@Entity
public class User {
  @Id @GeneratedValue
  private Long id;
  private String name;
  @Column(unique = true)
  private String email;
}

public interface UserRepo
  extends JpaRepository {
  Optional findByEmail(String e);
}
```

💡 Design entities carefully. Schema changes are expensive.

Phase 3: REST APIs

Day 4-5 — Core endpoints

```
@RestController
@RequestMapping("/api/v1/users")
public class UserController {

  @GetMapping("/{id}")
  public UserDTO getUser(@PathVariable Long id)

  @PostMapping
  public UserDTO createUser(@Valid @RequestBody)

  @PutMapping("/{id}")
  public UserDTO updateUser( ... )

  @DeleteMapping("/{id}")
  public void deleteUser( ... )
}
```

💡 Always version APIs: /api/v1/. Always use DTOs.

Phase 4: Security + Auth

Day 6-7 — Lock it down

```
1. Spring Security config
2. JWT token generation
3. JWT validation filter
4. BCrypt password encoding
5. Role-based access:
   @PreAuthorize("hasRole('ADMIN')")

6. CORS configuration
7. Rate limiting
```

💡 Security is not optional. Add it from day 1.

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Phase 5: Testing

Day 8-9 — Prove it works

```
Unit tests:
  @ExtendWith(MockitoExtension.class)
  → Mock dependencies
  → Test service logic

Integration tests:
  @SpringBootTest
  → Test full flow
  → Real DB (H2 in-memory)

API tests:
  @WebMvcTest
  → Test controllers
  → MockMvc requests
```

💡 No tests = no confidence. Write them.

Phase 6: Docker + Deploy

Day 10 — Ship it

```
1. Dockerfile
2. docker-compose.yml (app + DB)
3. Environment variables for secrets
4. GitHub Actions CI/CD:
   → Run tests
   → Build Docker image
   → Push to registry
   → Deploy to server

5. Spring Actuator for health checks
6. Logging with SLF4J
```

💡 If it's not deployed, it doesn't exist.

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Phase 7: Production Readiness

The final 10% that matters most

```
Checklist:
  ☐ Error handling (@ControllerAdvice)
  ☐ Input validation (@Valid)
  ☐ Pagination on list endpoints
  ☐ Database indexes on query columns
  ☐ API documentation (Swagger)
  ☐ Health endpoint (/actuator/health)
  ☐ Logging (request ID tracking)
  ☐ Monitoring (Prometheus + Grafana)
  ☐ README with setup instructions
```

💡 This checklist = what separates juniors from seniors

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```
done = true
```

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